

Analysis PhD Exam, August 2022

Write solutions in a neat and logical fashion, giving complete reasons for all steps. You must attempt SIX problems.

1. Let $(X, \mathcal{M})$ be a measurable space and ν a signed measure. Define the Hahn decomposition of X and the Jordan decomposition of ν . Prove that the Jordan decomposition is unique.
2. Let μ, ν be positive measures defined on the same measure space. Define what it means for ν to be absolutely continuous with respect to μ , and state the Radon–Nikodym theorem.
3. State Tonelli's theorem, and give an example to show that the σ -finiteness hypothesis is necessary.
4. Give an example of each of the following, if possible. (If not possible, give a brief explanation of why.) (When giving examples, be clear about what measure space you are working in.)
 - a) a sequence of functions converging in measure but not in L^1 ,
 - b) a sequence converging almost uniformly but not essentially uniformly,
 - c) a sequence converging in L^1 but not almost everywhere.
5. Let $(X, \mathcal{M}, \mu)$ be a σ -finite measure space. Prove that the simple functions which belong to $L^2(\mu)$ are dense in $L^2(\mu)$.
6. Let $\mathcal{X}$ be a Banach space and $\mathcal{N} \subset \mathcal{X}$ a closed subspace, and let $\pi : \mathcal{X} \rightarrow \mathcal{X}/\mathcal{N}$ be the quotient map. Show that if $f : \mathcal{X} \rightarrow \mathbb{F}$ is a bounded linear functional such that $f(n) = 0$ for all $n \in \mathcal{N}$, then there is a bounded linear functional $g : \mathcal{X}/\mathcal{N} \rightarrow \mathbb{F}$ such that $f = g \circ \pi$.
7. Let $\mathcal{X}$ be a reflexive Banach space, with dual space $\mathcal{X}^*$. Suppose $B : \mathcal{X} \times \mathcal{X}^* \rightarrow \mathbb{F}$ is a bilinear mapping, and there is a constant $C > 0$ so that

$$|B(x, x^*)| \leq C\|x\|\|x^*\| \text{ for all } x \in \mathcal{X}, x^* \in \mathcal{X}^*.$$

Prove that there is a bounded linear operator $T : \mathcal{X} \rightarrow \mathcal{X}$ such that $B(x, x^*) = x^*(Tx)$ for all $x \in \mathcal{X}, x^* \in \mathcal{X}^*$. Must T be unique?

8. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function with compact support, and let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be a nonnegative L^1 function with $\int_{-\infty}^{\infty} \phi(t) dt = 1$. For each real $\lambda > 0$, consider the function

$$f_\lambda(x) := \frac{1}{\lambda} \int_{-\infty}^{\infty} \phi\left(\frac{x-t}{\lambda}\right) f(t) dt.$$

Prove that $f_\lambda \rightarrow f$ uniformly on $\mathbb{R}$ as $\lambda \rightarrow 0$.